Chui Twin: Technical Appendix

Longaeva AI Hackathon, Stage 2 (October 2026). Code: https://github.com/Ben369369/Chui-Twin-Financial-Modelling

This appendix covers how the system works, how it was tested and where it falls short. It is meant for readers who want to check the claims. Full model math is in `docs/model.md`, and data sources and terms are in `provenance.md`.

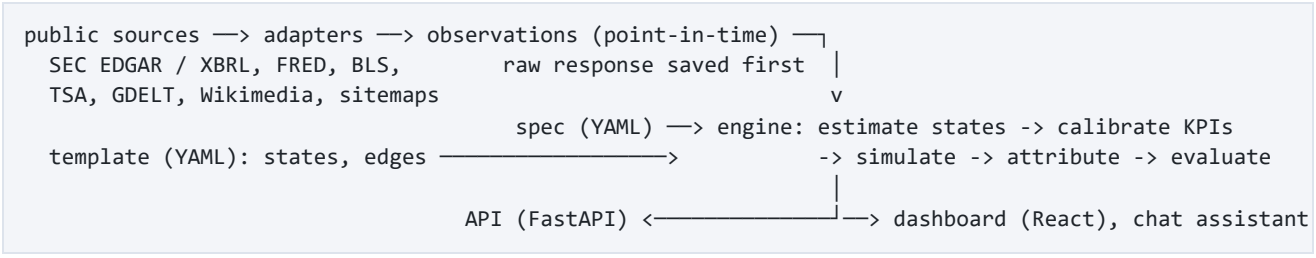
1. What the system does

Chui Twin turns a public company into a runnable operating model. Public data is treated as **noisy sensors of hidden operating states**: demand, capacity, price, input cost, labor, sentiment and expansion. One shared engine:

1. estimates those states each quarter,
2. links them to the KPIs the company reports through a causal graph,
3. simulates the next quarters with Monte Carlo,
4. traces every forecast back to the data behind it,
5. scores every twin the same way against simple and expert baselines.

The claim is about **the system, not one company**. Sixteen twins in five industries run on the same engine. Fourteen of them were built by a factory from the ticker alone, with no per-company tuning.

2. Architecture



Layer	Contents	What a new company or industry needs
`engine/`	estimation, calibration, simulation, attribution, evaluation, sealing	nothing (a test fails if a ticker, state name or series name appears here)
`adapters/`	SEC EDGAR and XBRL, FRED, BLS, TSA, Wikimedia, GDELT, a generic sitemap counter with Wayback history	usually nothing
`templates/`	an industry's states, default causal edges with priors, KPI structure	one YAML file per industry
`onboarding/`	ticker → SEC SIC code → industry pack → spec	the ticker
`specs/`	per company: KPIs, signals, edge overrides, extraction rules	written by hand (2) or by the factory (14)

Two tests enforce the separation. `test_engine_no_company_code.py` searches the engine for tickers, template state names and series prefixes. `test_template_agnostic.py` runs estimation and simulation end to end on a dummy template with made-up state names.

3. Data and point-in-time correctness

- **Coverage:** 155,913 observations of 64 signal series from 6 public sources (FRED, BLS, TSA, GDELT, Wikimedia, public sitemaps). From SEC filings there are 1,972 reported KPI values for 16 companies, plus 56 management-guidance values and 6 consensus values.
- **Raw first:** every fetch is saved untouched to `data/raw/<source>/<date>/` (270 snapshots) before parsing, so a parser can be fixed and rerun.
- **Point in time:** every observation stores `observed_at` (the period it describes) and `available_at` (when it could first be known). Release lags are set conservatively: 45 days after month end for monthly macro data, and the EDGAR acceptance time for filings. A backtest at time *t* uses only rows with `available_at ≤ t`. A test checks this.
- **Traceability:** every value carries a `source_ref`: the URL, the SEC accession number and exhibit, the raw snapshot path, and for text extractions the exact sentence it came from. The dashboard shows the sources behind each driver bar.
- **Legal data only:** robots.txt and terms were checked for every source, and nothing requires a login. Chipotle menu prices were not scraped, because they sit behind an API key in the site's JavaScript. Consensus estimates are hand-entered from public news quotes with the URL and quote kept.
- **Known caveat:** FRED and BLS serve the latest data vintage, so revised macro values can leak a little hindsight into old backtest quarters. Market prices and counts are not revised. A fix using ALFRED vintages is listed in `provenance.md`.

4. Model

Kept deliberately simple: about 30 scored quarters per twin cannot support a complex model.

1. **Signals → features:** each series becomes a decay-weighted level (half-life per signal), then a year-over-year change, standardized over the history available at forecast time.
2. **Features → states:** each state has a N(0, 1) prior. Each signal is a noisy reading of the state with precision p = c/(1 − c) · 0.5^(staleness/half-life), where *c* is the signal's confidence. The state mean is the precision-weighted average, so each signal's share of the state is exact and is stored.

3. **States → KPIs:** Bayesian linear regression with priors taken from the causal edges, in standardized units ("one sd of the parent moves the KPI by `prior_mean` of its sd"). Each KPI can be modeled as:
    - a **level**, with optional last-quarter and same-quarter-last-year terms;
    - a **change** since the last report;
    - an **anchored surprise** relative to the latest guidance or consensus published before the release.
4. **Identities and bounds:** accounting identities such as comps = transactions + average check are enforced on every simulated path by a minimum-variance projection. Physical bounds (a margin cannot exceed 100%) clip the paths, and the share clipped is reported.
5. **Simulation:** fully vectorized NumPy. States evolve as AR(1) processes with edge-propagated innovations, and scenario shocks decay with a half-life. A 10,000-path forecast takes about 0.03 s on a laptop, and a full refit about 0.06 s.
6. **Attribution:** a driver waterfall (each row linked to its sources) and leave-one-signal-out counterfactuals.

Shock half-lives for the scenario presets were measured from real events (`shocks/library.yaml`). For example, Chipotle's 2015 food-safety crisis decayed with a half-life of about 7.4 quarters, and COVID's hit to airlines and hotels with about 3.9.

5. Evaluation protocol

- **Walk-forward:** for each reported quarter, the forecast is made 1 hour before the earnings release, and everything is refit from scratch on data available at that moment. Tested from 2017Q1.
- **Excluded from scoring:** the COVID quarters (2020Q1–2021Q4), plus any regime break a spec declares (travel through 2022Q4). Specs can also leave those quarters out of calibration, and the travel specs do.
- **Baselines:** last quarter (`naive_last`), the same quarter last year (`seasonal_naive`), management guidance, and analyst consensus where public.
- **Metrics:** MAE, RMSE, bias, CRPS (scores the whole forecast distribution, not just the point) and 80% interval coverage. Skill = 1 − CRPS(twin)/CRPS(baseline) on the same quarters; positive means the twin is better.

6. Results

Full table: `docs/portfolio_scorecard.md` (16 twins, 33 KPI series).

Result	Value
KPI series beating "last quarter" on CRPS	**22 of 33**
KPI series beating "same quarter last year" on CRPS	**31 of 33**
DAL revenue growth vs Delta's own guidance (13 guided quarters)	skill **+0.28** (twin better)
CMG comps vs Chipotle's guidance (18 quarters)	skill **+0.41** (twin better)
CMG comps vs analyst consensus (6 quarters)	skill **−0.37** (consensus better)

Selected KPIs:

Twin	KPI	Quarters	MAE twin	MAE last quarter	Skill vs last quarter	80% coverage
DAL	revenue growth	26	2.72	4.15	+0.34	0.69
DAL	operating margin	26	2.39	5.03	+0.37	1.00
AAL (factory)	operating margin	26	2.60	5.68	+0.49	0.85
PHM (factory)	revenue growth	27	4.74	4.45	+0.33	0.96
CMG	comps	30	3.13	3.69	+0.21	0.80
CMG	revenue growth	30	3.86	3.92	−0.06	0.97

Reading the results honestly:

- Airlines are strongest, because TSA counts and jet fuel are timely, direct sensors.
- Restaurant and hotel revenue growth roughly ties "last quarter". Free public data carries little company-specific information there.
- Analyst consensus still beats the twin on Chipotle comps. Analysts see information that free public data does not.
- Interval coverage is uneven: some twins are too wide (coverage 1.00), and DAL revenue is a little narrow (0.69).
- **Selection bias:** the KPI structures for the two hand-built twins were chosen on the same backtest that is reported. The clean tests are (a) the 14 factory twins, which use industry defaults with no tuning, and (b) the sealed live forecasts below.

7. Live out-of-sample test: sealed forecasts

Before Delta reported Q3 2026, its forecast (revenue growth median 15.2%, 80% range 10.6 to 19.9, P(above guidance) 52%) was written to canonical JSON and its SHA-256 recorded. The hash was then timestamped through OpenTimestamps in Bitcoin block 969637 (Oct 2, 2026). Anyone can confirm that the forecast existed before the outcome, without trusting us.

`scripts/resolve_sealed.py`:

1. recomputes each hash,
2. accepts an outcome only if it became public after the seal,
3. scores the twin against guidance, last quarter and last year (`docs/sealed_results.md`).

Outcome (Delta reported Oct 9, 2026, 8-K accession 0000027904-26-000035). The hash verified.

KPI (GAAP)	Sealed median (80% range)	Actual	Inside 80%	Twin CRPS	Guidance	Last quarter	Last year
Operating margin	6.9 (−1.8 to 15.6)	**7.2**	yes	**1.60**	–	2.23	2.90
Revenue growth	15.2 (10.6 to 19.9)	**21.1**	no	3.93	6.07	**2.40**	14.72

- **Margin:** a hit. The twin was 0.3 points off and beat both naive baselines.
- **Revenue growth:** a miss. The actual was above the 80% range. The twin beat guidance but lost to "same as last quarter".

- **Cause:** GAAP revenue includes Delta's third-party refinery sales, which rose 76% with high fuel prices and added about 6.7 points of growth. The guidance the twin anchors on excludes the refinery; on that basis revenue grew 15.7%, 0.5 points from the twin's median. The twin anchored a GAAP KPI on non-GAAP guidance. The scored result stays as sealed. Future forecasts will match each anchor's definition to its KPI.

Chipotle's Q3 forecast will be sealed on Oct 27, before its Oct 28 report.

8. Validating extracted data: automation and manual review

`scripts/validate_ground_truth.py` runs automatic checks on **every** row (`docs/validation_report.md`):

Check	What it tests	Result (1,972 rows)
Text span	The source sentence appears verbatim in the saved filing, and its number (signed by its wording) equals the stored value	154 of 154 pass
Identity	comps = transactions + check on reported values, within the company's own rounding	14 of 14 pass
Bounds	Values inside physical limits	all pass
Outliers	Values more than 4 sd from the KPI's history, flagged for review	20 flagged: all real events (COVID, Starbucks' 2013 Kraft arbitration charge, Marriott's 2009 timeshare write-down, Pulte's 2009 Centex merger)

Building this caught three bugs **in the checker**, not in the data. Examples: "1.8% fewer transactions" was read as positive, and "December increased 14.7% and declined 4.8% for the full quarter" holds two numbers. That kind of variation in wording is why each rule is tested on real release text, and why every value keeps its source sentence.

Answer to "how much automation versus manual validation": automate every row, and send people a fixed-seed random sample of 20 (`docs/validation_sample.md`) plus anything flagged. Reviewers mark rows y or n, and `--apply` records them as verified. Reporting the agreement rate on the sample tells users how far to trust the automatic extraction.

9. Experiments, including negative results

`docs/experiments.md` records every signal tested. For example, JOLTS quits and job-openings rates were tested as hiring-difficulty proxies (the office-hours question on labor data) and did **not** improve the CMG or DAL margins. DAL operating margin MAE went from 2.43 to 2.49–2.61. They were not adopted. Industry-wide hiring conditions are not company-specific, and they move with wages, which the model already uses.

10. Scope

The system models **companies whose results come from observable operating drivers**: restaurants, retail, airlines, hotels and homebuilders today. It does not attempt the whole market:

- Banks need a new template (deposits, credit quality, margin).
- Binary-event companies such as clinical-stage biotech need a separate event engine.
- Both are on the roadmap in `docs/coverage_and_roadmap.md` and need no change to the existing engine.

For the next two weeks the priority is depth (validation, live sealed forecasts, LLM-assisted onboarding), not more companies.

11. Limitations

- **Short histories:** about 26–31 scored quarters per KPI. Skill differences under about 0.05 are noise.
- **A forecasting model, not a structural one:** scenario results are what the calibrated linear model implies. Moves far outside history (for example, fares up 30%, 2.5 sd) are extrapolation, and the dashboard and assistant say so.
- **Causal edges** between states come from priors, and are validated rather than discovered.
- **Seasonal margins** only tie the same-quarter-last-year baseline.
- **Rate limits:** the GDELT news-tone API rate-limits heavily, so some news-tone series can lag a few days. The daily job retries them.

12. Reproduce

```
pip install numpy "pydantic>=2" pyyaml requests fastapi uvicorn
python -m pytest -q                                    # 63 tests, no network (saved fixtures)
export CHUI_SEC_USER_AGENT="Your Name you@example.com"  # SEC requires a contact
python scripts/ingest.py specs/cmg.yaml --signals --ground-truth
python scripts/backtest.py specs/cmg.yaml --from 2017Q1 --exclude 2020Q1:2021Q4
python scripts/make_twin.py SBUX UAL MAR               # factory: ticker in, scored twin out
python scripts/validate_ground_truth.py
python scripts/resolve_sealed.py
uvicorn api.serve:app --port 8000                       # dashboard and API
```